ASSIGNMENT 2 – CO2 | EXECUTION EVIDENCE

Student: D. Saketh **Register No.:** 192525018
Environment: OpenJDK 21.0.12.1 (2026-08-18) **Compiler:** javac 21.0.12.1

The following output was captured from compiling and running the eight Java source files in the supplied project on this execution environment. Timing values are machine- and run-dependent; use them as this run's observations only, not universal performance claims.

Execution 1

Java execution evidence – StringComparison

```
Short ticket | repetitions=1000 | average ns/ticket: + 638.5, Builder 407.2, Buffer 428.7
50 fields | repetitions=1000 | average ns/ticket: + 9003.1, Builder 2647.4, Buffer 1169.1
200 fields | repetitions=500 | average ns/ticket: + 81718.0, Builder 3623.7, Buffer 3790.3
StringJoiner example: PNR:12345  | Name:Asha  | Coach:S1  | Fare:500
```

Execution 2

Java execution evidence – ListComparison

```
n=1000 | get(mid) ns: ArrayList=119370 LinkedList=776061 | front add/remove ns:
  ArrayList=63266 LinkedList=113159 | iteration ns: ArrayList=236757 LinkedList=315916
ArrayDeque Up Next sample: [2]
n=10000 | get(mid) ns: ArrayList=26249 LinkedList=1725563 | front add/remove ns:
  ArrayList=714499 LinkedList=53462 | iteration ns: ArrayList=629100 LinkedList=794366
ArrayDeque Up Next sample: [2]
n=50000 | get(mid) ns: ArrayList=41673 LinkedList=6324840 | front add/remove ns:
  ArrayList=1915611 LinkedList=74023 | iteration ns: ArrayList=2485236 LinkedList=2375214
ArrayDeque Up Next sample: [2]
```

Execution 3

Java execution evidence – Attendance

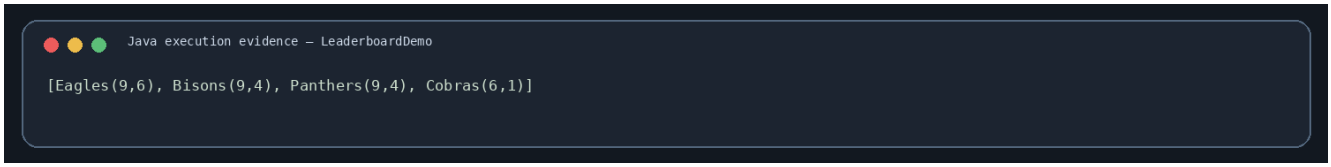
```
Unique students: 2
[192311001 Arun, 192311002 Divya]
```

Execution 4

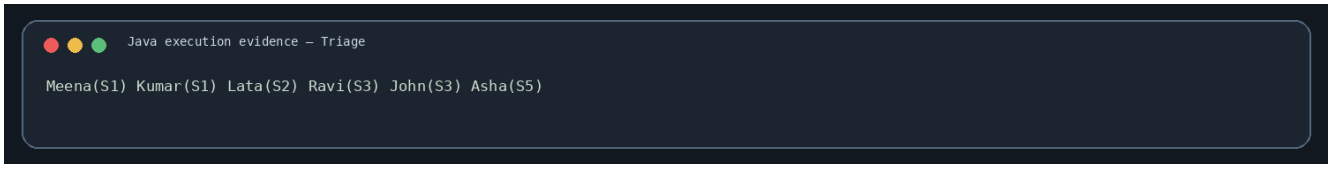
Java execution evidence – PharmacyMaps

```
HASH: 100000 lookups=12261987 ns, key-list report=2824861 ns, first=MED08850, last=MED01807,
  checksum=501000000
TREE: 100000 lookups=72462801 ns, key-list report=2404327 ns, first=MED00000, last=MED09999,
  checksum=501000000
LINKED: 100000 lookups=9779173 ns, key-list report=1377884 ns, first=MED00000,
  last=MED09999, checksum=501000000
```

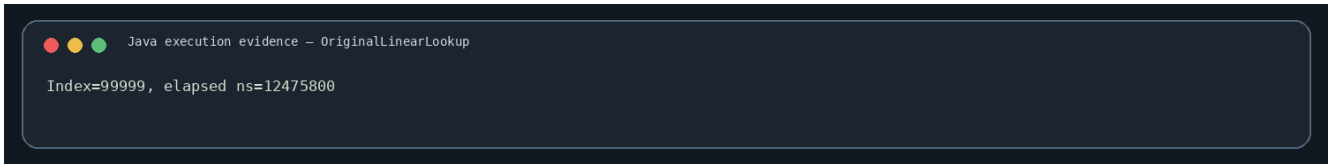
Execution 5



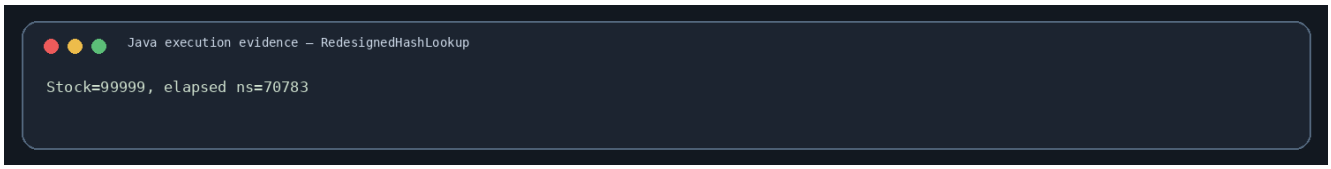
Execution 6



Execution 7



Execution 8



Complete captured console transcript

```
===== StringComparison =====
Short ticket | repetitions=1000 | average ns/ticket: + 638.5, Builder 407.2, Buffer 428.7
50 fields | repetitions=1000 | average ns/ticket: + 9003.1, Builder 2647.4, Buffer 1169.1
200 fields | repetitions=500 | average ns/ticket: + 81718.0, Builder 3623.7, Buffer 3790.3
StringJoiner example: PNR:12345  | Name:Asha  | Coach:S1  | Fare:500

===== ListComparison =====
n=1000 | get(mid) ns: ArrayList=119370 LinkedList=776061 | front add/remove ns: ArrayList=63266 LinkedList=113159 | it
ArrayDeque Up Next sample: [2]
n=10000 | get(mid) ns: ArrayList=26249 LinkedList=1725563 | front add/remove ns: ArrayList=714499 LinkedList=53462 | i
ArrayDeque Up Next sample: [2]
n=50000 | get(mid) ns: ArrayList=41673 LinkedList=6324840 | front add/remove ns: ArrayList=1915611 LinkedList=74023 |
ArrayDeque Up Next sample: [2]

===== Attendance =====
Unique students: 2
[192311001 Arun, 192311002 Divya]

===== PharmacyMaps =====
HASH: 100000 lookups=12261987 ns, key-list report=2824861 ns, first=MED08850, last=MED01807, checksum=501000000
TREE: 100000 lookups=72462801 ns, key-list report=2404327 ns, first=MED00000, last=MED09999, checksum=501000000
LINKED: 100000 lookups=9779173 ns, key-list report=1377884 ns, first=MED00000, last=MED09999, checksum=501000000

===== LeaderboardDemo =====
[Eagles(9,6), Bisons(9,4), Panthers(9,4), Cobras(6,1)]

===== Triage =====
Meena(S1) Kumar(S1) Lata(S2) Ravi(S3) John(S3) Asha(S5)

===== OriginalLinearLookup =====
Index=99999, elapsed ns=12475800

===== RedesignedHashLookup =====
Stock=99999, elapsed ns=70783
```